Week 11 Replica Method

Basic fact: $\forall z > 0$,

$$\lim_{k \to 0} \frac{1}{k}(z^k - 1) = \frac{d}{dz}(z^k - 1)\Big|_{z=0}$$

$k \uparrow 0^-$

$$= \frac{d}{dz}(e^{k \ln z} - 1)\Big|_{z=0}$$

$\mathbb{E}[z^{-1}]\ \mathbb{E}[z^{-2}] \cdots$

$$= \ln z$$

Example: E_1, E_2 random (energy) $\quad \Omega = \{1, 2\}$, $\mathbb{P}[\theta = 1] = \frac{e^{-\beta E_1}}{Z}$

partition function $\rightarrow$ $Z = e^{-\beta E_1} + e^{-\beta E_2}$ (β: inverse temperature)

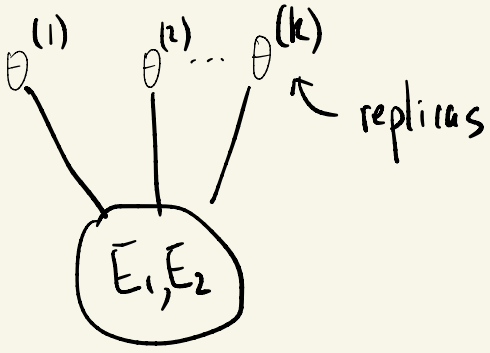
$$\mathbb{E}[Z^k] = \mathbb{E}\left[\sum_{\ell=0}^{k} \binom{k}{\ell} e^{-\ell\beta E_1 - (k-\ell)\beta E_2}\right]$$

$Z = \sum_{\theta_1 \cdots \theta_k} e^{-\beta \text{energy}}$

$= (\sum_{\theta_1} e^{-\beta E(\theta_1)})(\sum_{\theta_2} e^{-\beta E(\theta_2)})(\ \ )$

$= Z^k$

$$= \sum_{\ell=0}^{k} \binom{k}{\ell} \mathbb{E}\left[e^{-\ell\beta E_1 - (k-\ell)\beta E_2}\right] =: f(k)$$



$$\frac{d}{d\beta}\mathbb{E}[\log Z] = \frac{d}{d\beta}\mathbb{E}\left[\log \sum_{\theta} e^{-\beta E(\theta)}\right]$$

$$= \mathbb{E}\left[\frac{\sum_{\theta} -E(\theta) e^{-\beta E(\theta)}}{\sum_{\theta} e^{-\beta E(\theta)}}\right]$$

$$= \mathbb{E}_{\text{Gibbs}}[-E(\theta)]$$

$\beta \to \infty \longrightarrow \max\{-E(\theta)\}$

· A quick intro to large deviations

Example: $X_1 \cdots X_n$ iid $\mathrm{Ber}(1/2) =: Q$.

$\beta > 0$

Question: $\lim_{n\to\infty} \frac{1}{n} \ln \mathbb{E}\left[e^{-\beta \sum_{i=1}^n X_i}\right]$?

★ Method of types

empirical distribution of a sequence

$$\mathcal{P}_n := \left\{P : \exists\, x^n \in \{0,1\}^n,\ P = \hat{P}_{x^n}\right\}$$

$\hat{P}_{x^n}$ ↳ empirical distribution

$(0,1)\ \left(\frac{1}{n}, 1-\frac{1}{n}\right) \cdots \left(1-\frac{1}{n}, \frac{1}{n}\right),\ (1,0)$

$|\mathcal{P}_n| = n+1$ ↳ polynomial in n!

$$\mathbb{E}\left[e^{-\beta \sum_{i=1}^n X_i}\right] = \sum_{x^n} \mathbb{P}[X^n = x^n]\, e^{-\beta \sum_{i=1}^n x_i}$$

$$= \sum_{P \in \mathcal{P}_n} \left(\sum_{x^n : \hat{P}_{x^n} = P} \mathbb{P}[X^n = x^n]\, e^{-\beta \sum x_i} \right)$$

$\sum x_i \to n\mathbb{E}_P[X]$, X ↳ r.v. $\sim P$

$$= \sum_{P \in \mathcal{P}_n} \mathbb{P}\left[\hat{P}_{X^n} = P\right] e^{-\beta n \mathbb{E}_P[X]}$$

$\mathbb{P}\left[\hat{P}_{X^n} = P\right] = \exp\left(-n D(P\|Q) + o(n)\right)$

$$\doteq \sum_{P \in \mathcal{P}_n} \exp\left(-n D(P\|Q)\right) \cdot e^{-\beta n \mathbb{E}_P[X]}$$

$\doteq$: $= \exp(O(n))$ ...

Using the fact that $|P_n|$ is poly(n)

$$\mathbb{E}\left[e^{-\beta\sum_{i=1}^{n} x_i}\right] \doteq \max_{P\in P_n}\left\{\exp(-nD(P\|Q))\cdot e^{-\beta n E_P[X]}\right\}$$

By Continuity,

$$\lim_{n\to\infty}\frac{1}{n}\log \mathbb{E}\left[e^{-\beta\sum_{i=1}^{n} x_i}\right] = \max_{P\in\Delta}\left\{-D(P\|Q)-\beta E_P[X]\right\}.$$

↑ probability simplex; all possible distributions on $\{0,1\}$

- Least squares example.

$$A\in\mathbb{R}^{n\times p},\quad A_{ij}\sim\mathcal{N}(0,1),\quad \delta := \frac{n}{p} > 1$$

$$w\in\mathbb{R}^{n},\quad w_i\sim\mathcal{N}(0,n)$$

$$\hat{\theta} = \operatorname{argmin}_{\theta}\|w-A\theta\|$$

$\hat{\theta} = (A^TA)^{-1}A^Tw$ $\quad \mathbb{E}[\mathrm{Tr}(AA^T)^k]$

$\mathrm{Tr}[\mathrm{Cov}(\hat{\theta})] = n\,\mathbb{E}\left[\mathrm{Tr}\left((A^TA)^{-1}\right)\right] = np\int t^2 d\mu$

Goal show the $\hat{P}_{\hat{\theta}} \to \mathcal{N}(0,\frac{\delta}{\delta-1})$

- variance is $\frac{\delta}{\delta-1}$ $\mu : M_p$
- distribution is Gaussian

↳ $\{\mathbb{P}[\|\hat{\theta}\| - \mathbb{E}[\|\hat{\theta}\|] \geq t] \leq \cdots$
$Q\hat{\theta}$ dist $\hat{\theta}$ $\forall$ orthogonal Q

$$Z(\beta,n) := \int e^{-\frac{\beta}{2n}\|w-A\theta\|^2} d\theta$$

↳ $E(\theta)$

$$-\frac{\partial}{\partial\beta}\ln Z(\beta,n) = \frac{1}{Z(\beta,n)}\int E(\theta)\, e^{-\beta E(\theta)} d\theta$$

expectation of energy w.r.t. Gibbs measure

$Z(n) = \int_\theta e^{-\beta E(\theta) + \gamma\varphi(\theta)}$

$\frac{\partial}{\partial\gamma}\mathbb{E}[\ln Z]\Big|_{\gamma=0} = \mathbb{E}_{\theta\sim \text{Gibbs}}[-\varphi(\theta)]$

$\to \mathbb{E}_{N(0,\cdot)}[-\varphi)\ \forall\varphi$

→ Gibbs measure

↓ as $\beta\to\infty$

$\delta(\theta-\hat{\theta})$

Free energy:

$$F(\beta) = -\frac{1}{\beta}\ln Z(\beta,n)$$

Want: $\lim_{n\to\infty}\frac{1}{n}\mathbb{E}[F(\beta)] \neq -\frac{1}{n\beta}\ln\mathbb{E}[Z(\beta,n)]$

• $\mathbb{E}_{W,A}$

$$\mathbb{E}[Z^k] = \int \mathbb{E}_{W,A}\left[e^{-\frac{\beta}{2n}\sum_{a=1}^{k}\|W-A\theta^a\|^2}\right] d\theta^1\cdots d\theta^k$$

$$= \int \left(\mathbb{E}_{W,A} e^{-\frac{\beta}{2n}\sum_{a=1}^{k}(W_1 - A_1^T\theta^a)^2}\right)^n d\theta^1\cdots d\theta^k$$

$$= \int \det{}^{-n/2}\left(I+\beta E + \frac{\beta}{n}\sum_{j=1}^{p}\theta_j^{[k]}\theta_j^{[k]T}\right) d\theta^1\cdots d\theta^k$$

$Y\sim N(0,\Sigma)$

$\mathbb{E}[e^{-\frac{1}{2}\sum Y_i^2}] = \det{}^{-1/2}(I+\Sigma)$

where $\theta_j^{[k]} := \begin{bmatrix} \theta_j^1 \\ \vdots \\ \theta_j^k \end{bmatrix} \in \mathbb{R}^k$.

Define $Q := \frac{1}{p} \sum_{j=1}^{p} \theta_j^{[k]} \theta_j^{[k]T}$

μ : empirical distribution of $\{\theta_j^{[k]}\}_{j=1}^{p}$ (in $\mathbb{R}^k$)

$Q = \mathbb{E}_\mu[\theta\theta^T]$, $\theta \sim \mu$

- Differential entropy:

$h(\mu) := \mathbb{E}\left[\log \frac{1}{p(X)}\right]$, $X \sim \mu$, p is density of μ.

$$d\theta^1 \cdots d\theta^k = \exp\left\{ p \sup_{\mu : \mathbb{E}_\mu[\theta\theta^T] = Q} h(\mu) \right\} dQ$$

↳ Lebesgue measure on $\mathbb{R}^{k(k+1)/2}$

$$= (2\pi e)^{pk/2} \exp\left\{ \frac{p}{2} \log \det(Q) \right\} dQ$$

Exercise:

$$\max_{\mu : \mathbb{E}_\mu[X^2] \le \sigma^2} h(\mu) = \frac{1}{2} \log 2\pi e \sigma^2$$

Hint: $D(\mu \| N(0,1)) \ge 0$

· $p \to \infty$

$$\mathbb{E}[Z^k] = (2\pi e)^{pk/2} \int \det^{-n/2}\left(I + \beta E + \frac{\beta}{\delta} Q\right) \cdot \det^{p/2}(Q)\, dQ$$

$$\lim_{p\to\infty} \frac{1}{p} \log \mathbb{E}[Z^k] = \frac{k}{2}\log(2\pi e) + \sup_{Q \text{ is psd}} \left\{ -\frac{\delta}{2} \log\det\left(I + \beta E + \frac{\beta}{\delta} Q\right) + \frac{1}{2}\log\det(Q) \right\}$$

· Replica Symmetry Ansatz:

optimizing Q gives

$$Q = rI + qE = \begin{bmatrix} r+q & \cdots & q \\ q & \ddots & q \\ q & \cdots & r+q \end{bmatrix}$$

$\det(\lambda I + rE) = \lambda^{k-1}(\lambda + rk)$, $\mathbb{R}^{k\times k}$

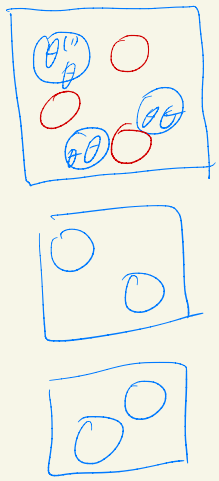
$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$

$q = \mathrm{Cov}(\theta_1^{(1)}, \theta_1^{(2)})$

$\theta^{(1)}$ $\theta^{(2)}$ ...

A, w

$\theta^{(2)}$ $\theta^{(1)}$



Continued

$$\Rightarrow \lim_{p\to\infty} \frac{1}{p} \ln \mathbb{E}[Z^k] = \sup_{r,q} \left\{ -\frac{(k-1)\delta}{2} \ln(1+\beta r/\delta) - \frac{\delta}{2}\ln(1+\beta/\delta + \beta q k/\delta + \beta k) + \frac{k-1}{2}\ln\frac{r}{\delta} + \frac{1}{2}\ln\frac{r+qk}{\delta} \right\} + \frac{k}{2}\ln(2\pi e)$$

Take $k \to 0$:

$$\lim_{p\to\infty} \frac{1}{p} \mathbb{E}[\ln Z]$$

$$= \lim_{p\to\infty} \frac{1}{p} \lim_{k\downarrow 0} \frac{1}{k} \ln \mathbb{E}[Z^k]$$

$$= \lim_{k\downarrow 0} \frac{1}{k} \lim_{p\to\infty} \frac{1}{p} \ln \mathbb{E}[Z^k]$$

$$= \sup_{q,r} \left\{ -\frac{\delta}{2} \ln(1+\beta r/\delta) - \frac{\beta q + \beta\delta}{2(1+\beta r/\delta)} + \frac{1}{2}\ln\frac{r}{q} + \frac{q}{2r} \right\} + \frac{1}{2}\ln(2\pi e)$$

Take $\frac{\partial}{\partial r}(\cdots) = 0$, $\frac{\partial}{\partial q}(\cdots) = 0$

$$\Rightarrow \quad \begin{cases} r = \dfrac{\delta}{\beta(\delta-1)} \\ q = \dfrac{\delta}{\delta-1} \end{cases}$$

$\beta \to \infty \Longrightarrow$ $r, q \to \dfrac{\delta}{\delta-1}$ as before!